

Graph Theory

Lemma 2.2.1 (Degree-sum Formula) *Let $G = (V, E)$ be a graph with m edges. Then $\sum_{v \in V} \deg(v) = 2m$.*

Lemma 2.2.2 *The number of odd degree vertices in a graph is an even number.*

respectively.

Let G be a graph with n vertices and m edges. Then by the degree-sum formula, the average degree of a vertex in G is $\frac{2m}{n}$ and hence $\delta(G) \leq \frac{2m}{n} \leq \Delta(G)$.



If $k = 2m$ is even, put all the vertices around a circle, and join each to its m nearest neighbors on either side.

25

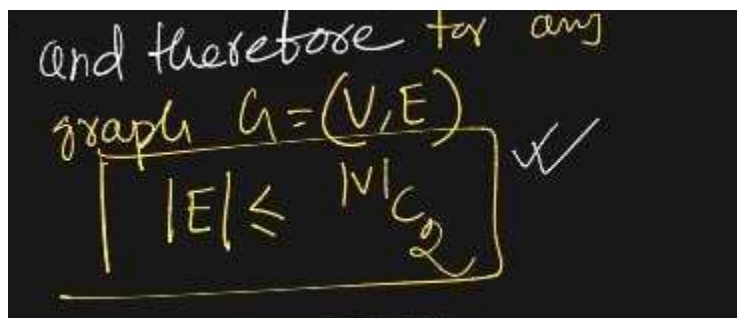


If $k = 2m + 1$ is odd, and n is even, put the vertices on a circle, join each to its m nearest neighbors on each side, and also to the vertex directly opposite.

Corollary 2.2.3 *Every regular graph with an odd degree has an even number of vertices.*

Corollary 2.2.4 *A k -regular graph with n vertices has $nk/2$ edges.*

vertices. Any graph is a subgraph of the complete graph with the same number of vertices and thus the number of edges in a graph with n vertices is at most $n(n-1)/2$.



Let graph G be isomorphic to H , and let $\overline{G}, \overline{H}$ denote their complements.

Since G is isomorphic to H , then there exists a bijection $f : V(G) \rightarrow V(H)$, such that $uv \in E(G)$ if and only if $f(u)f(v) \in E(H)$. -> [this should be edge set]

Equivalently, there exists a bijection $f : V(G) \rightarrow V(H)$, such that $uv \notin E(G)$ if and only if $f(u)f(v) \notin E(H)$. -> [this should be edge set]

Since the vertex set of G and $\overline{G}$ are the same, therefore f is a bijection from $V(\overline{G})$ to $V(\overline{H})$. Then suppose $uv \notin E(G)$, by definition of a complement, $uv \in E(\overline{G})$. Likewise, if $f(u)f(v) \notin E(H)$, then $f(u)f(v) \in E(\overline{H})$.

Hence $\overline{G}$ and $\overline{H}$ are isomorphic.

Induction

- 1) Equality
- 2) Inequality
- 3) x is divisible by y

$$\begin{aligned}
 H_{2^{k+1}} &= 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^k} + \frac{1}{2^k+1} + \cdots + \frac{1}{2^{k+1}} && \text{by the definition of harmonic number} \\
 &= H_{2^k} + \frac{1}{2^k+1} + \cdots + \frac{1}{2^{k+1}} && \text{by the definition of } 2^k\text{th harmonic number} \\
 &\geq \left(1 + \frac{k}{2}\right) + \frac{1}{2^k+1} + \cdots + \frac{1}{2^{k+1}} && \text{by the inductive hypothesis} \\
 &\geq \left(1 + \frac{k}{2}\right) + 2^k \cdot \frac{1}{2^{k+1}} && \text{because there are } 2^k \text{ terms each } \geq 1/2^{k+1} \\
 &\geq \left(1 + \frac{k}{2}\right) + \frac{1}{2} && \text{canceling a common factor of } 2^k \text{ in second term} \\
 &= 1 + \frac{k+1}{2}.
 \end{aligned}$$

Let $P(n)$ be the statement:

“the train stops at station n .”

We are given:

- The train stops at the first station.
- If the train stops at a station, then it also stops at the next station.

We will prove by induction that the train stops at every station.

Base case

For $n=1$:

The train stops at the first station.

So $P(1)$ is true.

Inductive step

Assume that the train stops at station k .

That means $P(k)$ is true.

We are given that if the train stops at a station, then it stops at the next station.

So if it stops at station k , it must also stop at station $k+1$.

Thus $P(k+1)$ is true.

Conclusion

Since $P(1)$ is true, and $P(k) \Rightarrow P(k+1)$, by mathematical induction $P(n)$ is true for all $n \in \mathbb{N}$.

Therefore, the train stops at **all stations**.

5. We proceed by induction. The basis step, $n = 0$, is true, since $1^2 = 1 \cdot 1 \cdot 3/3$. For the inductive step assume the inductive hypothesis that

$$1^2 + 3^2 + 5^2 + \cdots + (2k+1)^2 = \frac{(k+1)(2k+1)(2k+3)}{3}.$$

We want to show that

$$1^2 + 3^2 + 5^2 + \cdots + (2k+1)^2 + (2k+3)^2 = \frac{(k+2)(2k+3)(2k+5)}{3}$$

(the right-hand side is the same formula with $k+1$ plugged in for n). Now the left-hand side equals, by the inductive hypothesis, $(k+1)(2k+1)(2k+3)/3 + (2k+3)^2$. We need only do a bit of algebraic manipulation to get this expression into the desired form: factor out $(2k+3)/3$ and then factor the rest. In detail,

$$\begin{aligned} & (1^2 + 3^2 + 5^2 + \cdots + (2k+1)^2) + (2k+3)^2 \\ &= \frac{(k+1)(2k+1)(2k+3)}{3} + (2k+3)^2 \quad (\text{by the inductive hypothesis}) \\ &= \frac{2k+3}{3} ((k+1)(2k+1) + 3(2k+3)) = \frac{2k+3}{3} (2k^2 + 9k + 10) \\ &= \frac{2k+3}{3} ((k+2)(2k+5)) = \frac{(k+2)(2k+3)(2k+5)}{3}. \end{aligned}$$

7. Let $P(n)$ be the proposition $3 + 3 \cdot 5 + 3 \cdot 5^2 + \cdots + 3 \cdot 5^n = 3(5^{n+1} - 1)/4$. To prove that this is true for all nonnegative integers n , we proceed by mathematical induction. First we verify $P(0)$, namely that $3 = 3(5 - 1)/4$, which is certainly true. Next we assume that $P(k)$ is true and try to derive $P(k+1)$. Now $P(k+1)$ is the formula

$$3 + 3 \cdot 5 + 3 \cdot 5^2 + \cdots + 3 \cdot 5^k + 3 \cdot 5^{k+1} = \frac{3(5^{k+2} - 1)}{4}.$$

All but the last term of the left-hand side of this equation is exactly the left-hand side of $P(k)$, so by the inductive hypothesis, it equals $3(5^{k+1} - 1)/4$. Thus we have

$$\begin{aligned} 3 + 3 \cdot 5 + 3 \cdot 5^2 + \cdots + 3 \cdot 5^k + 3 \cdot 5^{k+1} &= \frac{3(5^{k+1} - 1)}{4} + 3 \cdot 5^{k+1} \\ &= 5^{k+1} \left(\frac{3}{4} + 3 \right) - \frac{3}{4} = 5^{k+1} \cdot \frac{15}{4} - \frac{3}{4} \\ &= 5^{k+2} \cdot \frac{3}{4} - \frac{3}{4} = \frac{3(5^{k+2} - 1)}{4}. \end{aligned}$$

9. a) We can obtain a formula for the sum of the first n even positive integers from the formula for the sum of the first n positive integers, since $2 + 4 + 6 + \cdots + 2n = 2(1 + 2 + 3 + \cdots + n)$. Therefore, using the result of Example 1, the sum of the first n even positive integers is $2(n(n+1)/2) = n(n+1)$.
b) We want to prove the proposition $P(n) : 2 + 4 + 6 + \cdots + 2n = n(n+1)$. The basis step, $n = 1$, says that $2 = 1 \cdot (1 + 1)$, which is certainly true. For the inductive step, we assume that $P(k)$ is true, namely that

$$2 + 4 + 6 + \cdots + 2k = k(k+1),$$

and try to prove from this assumption that $P(k+1)$ is true, namely that

$$2 + 4 + 6 + \cdots + 2k + 2(k+1) = (k+1)(k+2).$$

(Note that the left-hand side consists of the sum of the first $k+1$ even positive integers.) We have

$$\begin{aligned} 2 + 4 + 6 + \cdots + 2k + 2(k+1) &= (2 + 4 + 6 + \cdots + 2k) + 2(k+1) \\ &= k(k+1) + 2(k+1) \quad (\text{by the inductive hypothesis}) \\ &= (k+1)(k+2), \end{aligned}$$

as desired, and our proof by mathematical induction is complete.

15. The base case of the statement $P(n) : 1 \cdot 2 + 2 \cdot 3 + \cdots + n(n+1) = n(n+1)(n+2)/3$, when $n = 1$, is $1 \cdot 2 = 1 \cdot 2 \cdot 3/3$, which is certainly true. We assume the inductive hypothesis $P(k)$, and try to derive $P(k+1)$:

$$1 \cdot 2 + 2 \cdot 3 + \cdots + k(k+1) + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$$

Starting with the left-hand side of $P(k+1)$, we have

$$\begin{aligned} (1 \cdot 2 + 2 \cdot 3 + \cdots + k(k+1)) + (k+1)(k+2) &= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) \quad (\text{by the inductive hypothesis}) \\ &= (k+1)(k+2) \left(\frac{k}{3} + 1 \right) = \frac{(k+1)(k+2)(k+3)}{3}, \end{aligned}$$

the right-hand side of $P(k+1)$.